2024 6th International Conference on Frontier Technologies of Information and Computer (ICFTIC) | 979-8-3315-4175-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICFTIC64248.2024.10913101

An arXiv Paper Question-Answering System Based on Qwen and RAG

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***Abstract*—This paper explores combining large language models and retrieval-augmented generation (RAG) techniques to build a question-answering system for academic papers. The system is built upon Qwen2.5 models and open-source tools like LlamaIndex, utilizing the arXiv dataset to implement an end-to-end pipeline from user queries to relevant answers through models including query routing, hybrid retrieval, and answer generation. In the retrieval phase, the system leverages both the BGE-M3 embedding model and the BGE-Reranker-V2-M3 reranking model. The generation stage integrates knowledge from multiple sources using a fine-tuned Qwen2.5 model. Experiments show significant improvements in retrieval accuracy and answer coherence by incorporating Reranker and RAG-Fusion. This study highlights the potential of LLM and RAG technologies in academic applications and offers insights for enhancing paper question-answering systems.**

Keywords—Large language models; question-answering system; qwen; retrieval-augmented generation

I. Introduction

The number of academic papers is experiencing a period of rapid growth. According to arXiv.org, over 100,000 new papers were added to the computer science(CS) category in the first 11 months of 2024 alone. This massive volume of literature presents a significant challenge for researchers trying to accurately and efficiently locate the information they need. Traditional keyword-based search methods struggle to deeply understand the content of papers, often failing to deliver relevant and comprehensive results. There is an urgent need for new technologies that can intelligently assist researchers in effectively utilizing these resources.

Recent years have witnessed breakthroughs in natural language understanding and generation with Large Language Models (LLMs). LLMs can learn rich semantic knowledge from massive text corpora, demonstrating powerful capabilities in language comprehension and reasoning. Applying LLMs to the academic domain promises to enable deeper semantic analysis of papers, thereby improving the quality of scholarly knowledge services, despite several key limitations as illustrated in Fig. 1.

However, directly applying LLMs to lengthy academic papers presents challenges. LLMs have limitations in handling long contexts[1], and even long-context LLMs may struggle to accurately grasp the key points of an article (as seen in the open-source GitHub project "Needle In A Haystack - Pressure Testing LLMs")[1]. Furthermore, inference speed can be slow[2]. These challenges motivate us to combine LLMs with Retrieval-Augmented Generation (RAG) techniques to efficiently process academic papers.

This paper explores the combination of LLMs and RAG techniques to build an efficient semantic search and question-answering system for academic papers. The significance of this research lies in several aspects:

Enhanced Research Efficiency: The paper question-answering system can help researchers quickly extract the information they need from a vast number of papers, significantly improving research efficiency. This has important implications for accelerating scientific innovation and promoting interdisciplinary research.

Noval Integration of LLMs and RAG: This work explores the innovative application of LLMs and RAG in processing academic papers, enriching the practical cases of large models in specialized domains. The modular design of the system offers a degree of generalizability and can provide a reference for intelligent solutions in other long document scenarios.

Efficiency with Smaller LLMs: Instead of relying on large-scale LLMs, this paper adopts a smaller 7B model with fine-tuning and RAG. This achieves understanding and question-answering capabilities within limited computational resources, providing a viable approach for small and medium-sized institutions to utilize LLMs for academic services.

Enhanced Dataset: This work enriches an open-source academic paper dataset by supplementing missing fields, increasing its richness and laying a data foundation for further research on academic large language models.

Project open-source repository: https://github.com/gxmzuai/arXiver-Qwen.

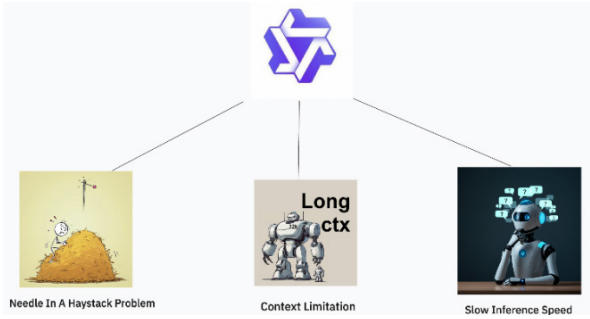

Figure 1. LLM limitations in handling long contexts

[1] https://github.com/gkamradt/LLMTest_NeedleInAHaystack

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II. Related Work

A. Large Language Models

Large language models (LLMs) represent a significant breakthrough in natural language processing. In recent years, with the continuous increase in model scale and the enrichment of pre-training corpora, LLMs have demonstrated remarkable zero-shot and few-shot learning capabilities.

As shown in Fig. 2, Qwen2.5[3] is a series of multilingual, multi-domain foundation models recently developed by Alibaba, including the general-purpose language model Qwen2.5, the code model Qwen2.5-Coder, and the mathematics model Qwen2.5-Math. It represents a significant improvement over its predecessor, Qwen2[4]. Qwen2.5 is pre-trained on a massive corpus of 18 trillion tokens and excels in knowledge acquisition, programming, and mathematics, making it a leader among current open-source LLMs.

Qwen2.5-coder[5] is a model series built upon Qwen2.5, specifically pre-trained and fine-tuned for programming tasks. Through data cleaning and balanced data mixing, Qwen2.5-coder retains general and mathematical skills while demonstrating outstanding code generation capabilities. It achieves state-of-the-art (SOTA) performance among open-source LLMs in over 10 benchmark tests for code generation, completion, reasoning, and repair.

QwQ-32B-Preview[6] is an experimental research model released by the Qwen team on November 28, 2024. Leveraging Chain-of-Thought(CoT)technology[7], this model exhibits enhanced comprehension of complex problems and deeper reasoning capabilities, providing an open-source alternative with reasoning capabilities on par with top-tier models like those in OpenAI's o1 category.

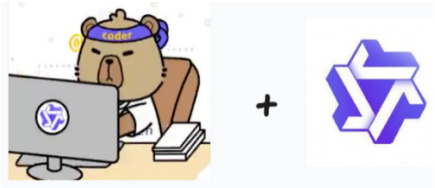

Figure 2. Qwen series lead in open-source LLMs

B. Retrieval-Augmented Generation

To address the limitations of LLMs in knowledge acquisition when applied to complex tasks, Retrieval-Augmented Generation (RAG) has emerged. The core idea of RAG is to incorporate external knowledge bases during the generation process to enhance the knowledge access and reasoning abilities of LLMs.

Keyword Retrieval: RAG's keyword retrieval finds relevant documents by matching keywords in user queries with keywords in documents. For example, when you type "Alibaba" into a search engine, it returns web pages containing the word "Alibaba."

Semantic Retrieval: RAG's semantic retrieval is a search method based on semantic understanding. It aims to understand the meaning of both user queries and documents, retrieving relevant documents based on semantic similarity. For example, when you search for "Alibaba," semantic retrieval will not only return web pages containing "Alibaba" but also pages containing semantically related terms like "Jack Ma" or "Tongyi Qianwen."

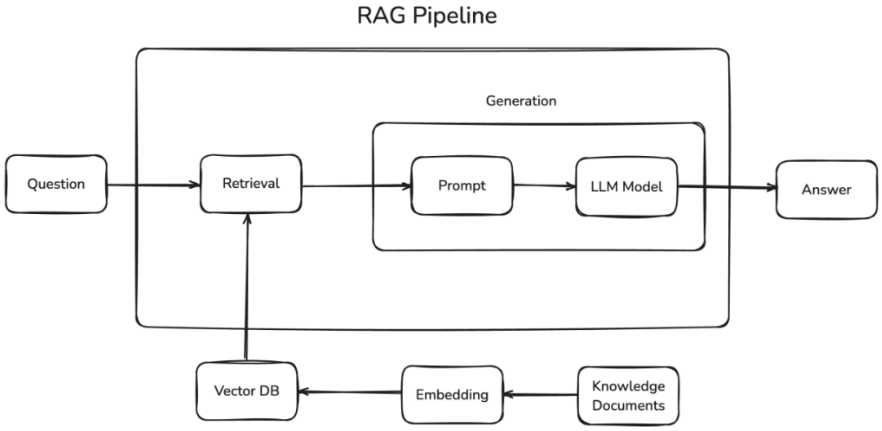

Figure 3. A naive RAG semantic search pipeline

Fig.3 illustrates a simplified RAG semantic retrieval process. Embedding data from documents is stored in a vector database. When a user submits a query, the vector database is searched to retrieve semantically similar passages. These passages are then added to a prompt template and sent to the LLM for response generation.

Here are some recent research advances in the field of RAG:

- **RAG-Fusion[8]:** This approach combines RAG with Reciprocal Rank Fusion(RRF) [9]. By generating multiple sets of queries and re-ranking them using reciprocal rank scores, it fuses documents and scores to optimize RAG performance. Experimental results demonstrate that RAG-Fusion can answer questions from multiple perspectives, providing more comprehensive and accurate answers.

- **Self-RAG[10]:** To further improve the quality and factuality of LLMs, Self-RAG introduces a novel self-reflective RAG framework. This framework trains models to dynamically retrieve passages based on demand and introduces "reflection tokens" to reflect on the content of retrieved passages and the model's own generated output. The use of reflection tokens makes the model more controllable during inference, allowing it

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to flexibly adjust its behavior according to the needs of different tasks. Experimental results show that Self-RAG significantly surpasses existing SOTA LLMs and RAG models in open-domain question-answering, reasoning, and factual verification tasks.

- **CRAG[11]:** Corrective Retrieval Augmented Generation (CRAG) is an innovative RAG framework designed to improve the robustness of generative LLMs. While RAG methods can effectively provide LLMs with necessary external knowledge, their performance heavily relies on the relevance of retrieved documents. To address this, CRAG introduces a lightweight retrieval evaluator to assess the overall quality of query-relevant documents and triggers different knowledge retrieval operations based on confidence. Experiments on four datasets covering short-text and long-text generation tasks demonstrate that CRAG significantly enhances the performance of RAG-based methods.

- **Adaptive-RAG[12]:** This adaptive question-answering framework dynamically selects strategies from simplest to most complex based on query complexity. This selection process is operated by a classifier, which is a smaller trained language model that predicts the complexity of incoming queries based on the model's own prediction and automatically collected labels derived from inherent inductive biases in the dataset. This method strikes a balance between iterative RAG and single-step RAG, as well as no-retrieval methods, to respond to various query complexities.

C. *Application Research in Academia*

Introducing LLMs and RAG into the academic domain promises to enable intelligent processing of vast scientific literature, significantly improving research efficiency. However, academic papers are characterized by high specialization, length, and complex structures, posing higher demands on models' knowledge representation and reasoning abilities. Below are recent advancements in RAG applications in academia.

- **PaperQA[13]:** Lála et al. proposed the PaperQA system, which can perform information retrieval within full-text scientific articles, assess the relevance of sources and passages, and provide answers using RAG. Compared to existing LLMs and human researchers, PaperQA demonstrates superior performance. Additionally, they introduced the LitQA benchmark to evaluate the system's capabilities on complex scientific literature research tasks. Experimental results show that PaperQA performs comparably to expert human researchers on LitQA. The authors believe that the development of systems like PaperQA will drive innovation in scientific research at lower cost and faster speed.

- **PaperQA2[14]:** Skarlinski et al. further developed the PaperQA2 system and the LitQA2 benchmark. PaperQA2 performs on par with or better than experts on literature research tasks and generates Wikipedia-style summaries that are more accurate than existing human-written summaries. Moreover, PaperQA2 can identify contradictions in scientific literature at scale, a task that is extremely challenging for human experts. This research demonstrates that language models can now surpass domain experts in challenging scientific tasks.

- **OpenScholar[15]:** This is a recent RAG system specifically designed for scientific literature. It identifies relevant passages from 45 million open-access papers and generates answers with citation support. Researchers developed a large-scale benchmark, ScholarQABench, covering multiple disciplines to evaluate OpenScholar. Results show that despite its smaller size, OpenScholar-8B outperforms GPT-4o and PaperQA2 in terms of correctness. OpenScholar exhibits almost no citation hallucinations, which occur in 78-90% of GPT-4o's outputs, and its citation accuracy is comparable to human experts. Answers generated by OpenScholar are also preferred over those written by human experts.

In conclusion, LLMs and RAG demonstrate great potential in the academic domain, promising to become valuable assistants for researchers and promote the efficient utilization and innovation of scientific knowledge. This work will leverage the arXiv dataset to enrich their practical applications within academia.

III. System Architecture

The overall architecture of the paper question-answering system is shown in Fig. 4. The system adopts a modular design approach and consists of four core modules: data processing, routing selection, content retrieval, and answer generation. Each module works collaboratively in a pipeline manner, ensuring the coherence of the end-to-end process from user query to answer generation while achieving task decoupling and modularity.

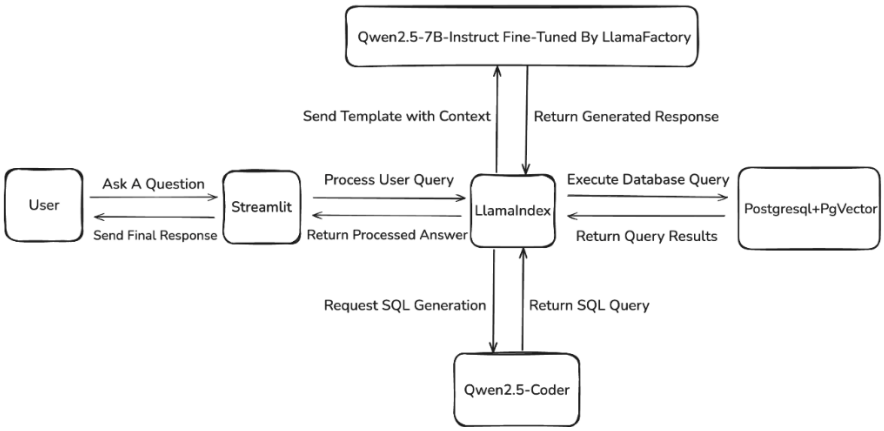

Figure 4. System Architecture

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A. Data Processing

The foundation of the data for the paper question-answering system is the open-source Arxiver dataset [2] available on HuggingFace. This dataset includes 63,357 arXiv papers published between January and October 2023. It covers the original arXiv ID, titles, abstracts, authors, publication dates, links, and the full Markdown text of the papers.

To efficiently manage and retrieve the paper data, we designed two tables, “papers” and “paper_chunks” in a PostgreSQL database, as shown in Table I and Table II. The “papers” table stores the original paper metadata.

To enrich the data dimensions, we supplemented two fields, “primary_category” and “categories” by requesting arXiv's free API at a reasonable request rate. This provides more possibilities for subsequent retrieval and analysis.

To enable efficient keyword fuzzy matching, we implemented a GIN index using pg_trgm on the text fields of the “papers” table, specifically “title” and “abstract.” Simultaneously, we established a B-tree index on numerical and categorical fields like “published_date” to optimize the performance of range queries and exact matches.

TABLE I. PAPERS TABLE

Field Name	Data Type	Description
id	VARCHAR(50)	Paper ID
title	TEXT	Paper title
abstract	TEXT	Paper abstract
authors	TEXT	paper authors list
published_date	TIMESTAMPTZ	Paper publication date
link	TEXT	Paper link
markdown	TEXT	Paper content
created_at	TIMESTAMPTZ	Creation timestamp
primary_category	VARCHAR(20)	Main category
categories	TEXT[]	Paper categories

The "paper_chunks" table stores the segments of the full paper text and their corresponding semantic vectors. We utilized the MarkdownNodeParser provided by the LlamaIndex framework to parse the full Markdown text. This parser intelligently chunks the text based on the heading hierarchy of Markdown, effectively maintaining the semantic integrity of the document. The “include_prev_next_rel” parameter of the parser is used to maintain the relationship between nodes, mitigating context loss caused by chunking.

The embedding model employed is the BGE-M3 model[16] open-sourced by the Beijing Academy of Artificial Intelligence (BAAI). This model supports cross-lingual embeddings, enabling retrieval of semantically similar English chunks in response to Chinese queries. Each chunk of the paper is converted into a 1024-dimensional vector by the BGE-M3 model. We select the corresponding vectors for dense retrieval and store them along with the original chunk content in the “paper_chunks” table for subsequent semantic retrieval.

“(paper_id, chunk_index)” serves as the composite primary key of the “paper_chunks” table. An HNSW index is built on the embedding vector field. Utilizing the functions provided by the PgVector extension, we achieve efficient approximate nearest neighbor vector search.

By comprehensively utilizing the “papers” and “paper_chunks” tables, the system can support fast keyword retrieval while enabling semantic similarity-based retrieval, providing solid data support for subsequent content retrieval and answer generation.

TABLE II. PAPER CHUNKS TABLE

Field Name	Data Type	Description
paper_id	VARCHAR(50)	Paper ID
chunk_index	INTEGER	Chunk index
content	TEXT	Chunk text
embedding	VECTOR(1024)	Chunk embedding
created_at	TIMESTAMPTZ	Creation timestamp

B. Routing Selection

The primary motivation for introducing the routing selection module is to improve the overall efficiency of the paper question-answering system. By filtering out questions irrelevant to the system, this module can guide users to focus on questions within the domain of academic papers. Simultaneously, it dynamically selects semantic retrieval or keyword retrieval strategies for specific types of queries, further enhancing retrieval quality and the final response quality. This also reduces the latency of the RAG process.

As illustrated in Fig. 5, the basic routing selection process is as follows: First, the user inputs a question, and the Qwen model makes a preliminary judgment on whether the question belongs to the academic paper domain. If it is identified as an irrelevant question, the system prompts the user to rephrase the question until it falls within the scope of academic papers. Once confirmed as a paper-related question, the query enters the RAG process, where it is further determined whether to trigger PostgreSQL keyword retrieval or PostgreSQL + PgVector semantic retrieval.

C. Content Retrieval

The content retrieval module shoulders the responsibility of finding the most relevant passages from a vast number of papers to the user's query. It is a crucial link in providing knowledge sources for answer generation. To balance retrieval flexibility and accuracy, we adopt a hybrid strategy that combines keyword matching and semantic retrieval.

First, keyword retrieval relies on the LLM to generate standardized SQL query statements. Specifically, we concatenate the user input with a predefined keyword retrieval SQL template to form a prompt, which is input to the current

[2] https://huggingface.co/datasets/neuralwork/arxiver

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SOTA open-source LLM for code—Qwen2.5-Coder. Due to its outstanding performance on code tasks, Qwen2.5-Coder can automatically generate safe and efficient SQL statements based on user intent and context, ensuring the targeted and reliable nature of keyword retrieval.

However, keyword matching often struggles with texts that are semantically related but have significant lexical differences. To overcome this limitation, we further introduce semantic retrieval. First, the BGE-M3 model encodes the user query into a 1024-dimensional semantic vector. Then, cosine similarity is calculated on the "embedding" field of the "paper_chunks" table to identify the top-10 paper segments closest to the user query in the vector space.

As keyword matching and semantic retrieval yield candidate sets of varying relevance and quality, a reranking step is crucial for selecting the most appropriate segments. To select the most matching and high-quality segments, we introduce the BGE-Reranker-V2-M3 model into the ranking stage. This model accepts (query, document) pairs and outputs the relevance score of the document relative to the query. By pairing the query with each candidate segment and having the Reranker predict relevance and re-rank based on the score, we can more comprehensively evaluate the matching degree between the query and the segment, compensating for the limitations of a single semantic dimension. Finally, the top-5 ranked candidate segments are used as relevant knowledge sources and input to the subsequent answer generation module.

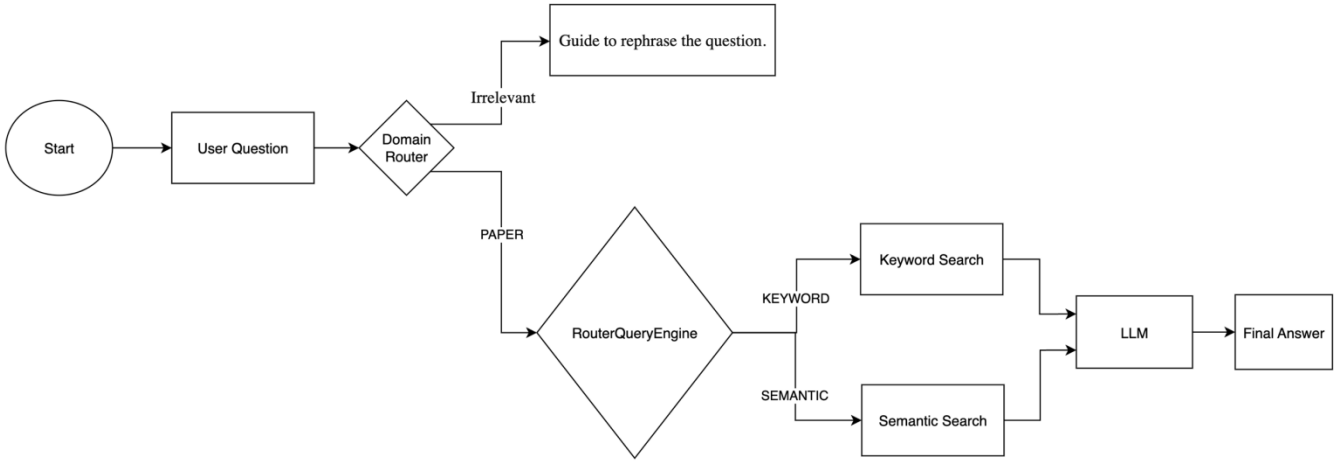

Figure 5. Routing Selection, Content Retrieval and Answer Generation Flowcart

D. Answer Generation

The answer generation module, based on the paper information obtained through content retrieval, aims to produce coherent and natural answer texts. We selected the Qwen2.5-7B-Instruct model, fine-tuned using LLaMAFactory[17] for enhanced cognitive capabilities, to address the paper question-answering scenario. To fully utilize the retrieved knowledge, we incorporate relevant information along with the questions into a predefined prompt template to form contextual information. The Qwen model then generates responses based on this enriched background.

By fusing multi-source information and optimizing through query routing, hybrid retrieval, and ranking, the Qwen model generates high-quality answers that are accurate, informative, and logically structured.

IV. Experiments and Analysis

A. Experimental Design

To evaluate the effectiveness of the paper question-answering system in both retrieval and generation aspects, we automatically constructed a dataset of 300 question-answer pairs for semantic retrieval and manually curated a dataset of 50 question-answer pairs for keyword retrieval.

We constructed 300 question-answer pairs for semantic retrieval, leveraging Google's Gemini 1.5 Flash model. From each of the 149 paper categories in the Arxiver dataset, we first identified two high-quality text chunks. These, along with two additional randomly selected chunks, were then used to generate question-answer pairs. To ensure quality, each chunk was evaluated beforehand by Gemini 1.5 Flash for its suitability in generating meaningful question-answer pairs based on a predefined prompt template.

The 50 question-answer pairs for keyword retrieval were manually constructed to primarily examine the system's ability to retrieve and respond to queries regarding paper metadata (such as paper title, abstract, and author information).

Our experimental baseline was the Qwen2.5 series models with a conventional RAG (without Reranker) pipeline. We then conducted comparative experiments by adding a Reranker and incorporating other RAG techniques such as RAG-Fusion and Self-RAG.

After incorporating the Reranker and RAG-Fusion modules, we employed a two-stage process to determine document relevance. First, RAG-Fusion's RRF (Reciprocal Rank Fusion) algorithm processed the documents, followed by the Reranker, which identified the top-5 most relevant segments. The final document relevance score was calculated as a weighted average, with the Reranker score contributing 0.7 and the RAG-Fusion RRF score contributing 0.3.

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$$\text{Relevance Score} = 0.7 \times \text{Rerank Score} + 0.3 \times \text{RRF Score} \quad (1)$$

B. Experimental Results and Analysis

The semantic retrieval evaluation primarily assessed the performance of the paper question-answering system from two aspects: retrieval quality and generated answer quality.

Retrieval quality was measured using the following metrics:

a. Precision with Positional Weights: Traditional precision only considers the proportion of relevant documents, while users often pay more attention to the top-ranked results. Therefore, we introduced precision with positional weights, calculated as the ratio of the sum of positional weights of relevant documents (with semantic similarity greater than a similarity threshold) to the sum of all positional weights.

$$\text{Weighted Precision} = \frac{\sum_{i=1}^{n} w_i \cdot I(sim_i)}{\sum_{i=1}^{n} w_i} \quad (2)$$

$$w_i = \frac{1}{\log_2(i+2)} \quad (3)$$

$$I(sim_i) = \begin{cases} 1, & \text{if } sim_i > \theta, \\ 0, & \text{if } sim_i \le \theta. \end{cases} \quad (4)$$

b. Recall: This is calculated as the ratio of the number of documents with semantic similarity greater than a similarity threshold to the total number of documents.

$$\text{Semantic Recall} = \frac{\sum_{i=1}^{n} I(sim_i)}{n} \quad (5)$$

$$I(sim_i) = \begin{cases} 1, & \text{if } sim_i > \theta, \\ 0, & \text{if } sim_i \le \theta. \end{cases} \quad (6)$$

c. Mean Reciprocal Rank (MRR): This is the reciprocal of the rank of the first correct answer chunk. A higher MRR indicates that the correct information is ranked higher.

$$\text{MRR} = \frac{1}{\text{rank}_{\text{first}}} \quad (7)$$

$$\text{rank}_{\text{first}} = \min\{i \mid \text{chunk}_i = \text{expected chunk}\} \quad (8)$$

d. Normalized Discounted Cumulative Gain (NDCG): This is an evaluation metric that comprehensively considers both relevance strength and ranking position, calculated as follows:

d1. Discounted Cumulative Gain (DCG): The relevance score of each position is weighted and summed, with documents at lower positions receiving smaller weights, reflecting the importance of position.

$$\text{DCG} = \text{rel}_1 + \sum_{i=2}^{n} \frac{\text{rel}_i}{\log_2(i+1)} \quad (9)$$

d2. Ideal Discounted Cumulative Gain (IDCG): The DCG value is calculated for the ideal ranking, i.e., the DCG value after sorting all documents by relevance score from highest to lowest. This represents the optimal ranking effect that can be achieved for the current query.

$$\text{IDCG} = \text{rel}_{1'} + \sum_{i=2}^{n} \frac{\text{rel}_{i'}}{\log_2(i+1)} \quad (10)$$

d3. Normalization: Finally, DCG is divided by IDCG for normalization, resulting in an NDCG score between 0 and 1. An NDCG closer to 1 indicates that the ranking of the actual retrieval results is closer to the ideal ranking, meaning that documents with high relevance are ranked higher; conversely, an NDCG closer to 0 indicates poorer ranking quality.

$$\text{NDCG} = \frac{\text{DCG}}{\text{IDCG}} \quad (11)$$

Generated answer quality was measured using the following metrics:

Faithfulness: Whether the generated answer faithfully originates from the retrieved results, without fabricated or irrelevant information.

Relevancy: Whether the content of the answer is relevant and highly related to the original question.

Both faithfulness and relevancy were scored by an LLM (using OpenAI's o1-mini model, as the Qwen QwQ-32B-Preview model had a certain probability of triggering a refusal to respond, so we opted for a powerful proprietary model as a substitute) with a value of 0 or 1, where a score of 1 indicates good generation quality.

By calculating the average of each metric on the test set, we can quantitatively evaluate the overall performance under different system configurations, providing a reference for further optimization.

The experimental results for semantic retrieval evaluation are presented in Fig. 6 and Fig. 7.

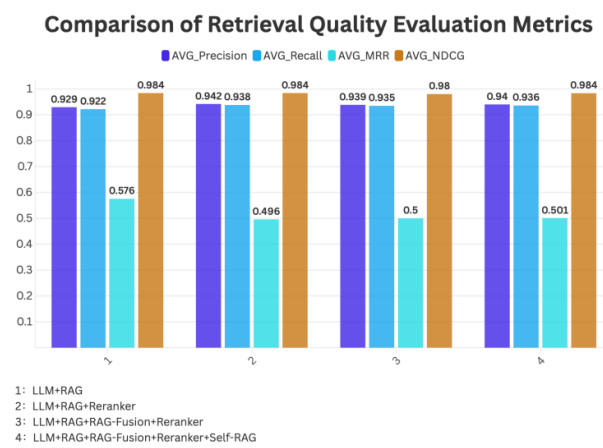

Figure 6. Comparison of Retrieval Quality

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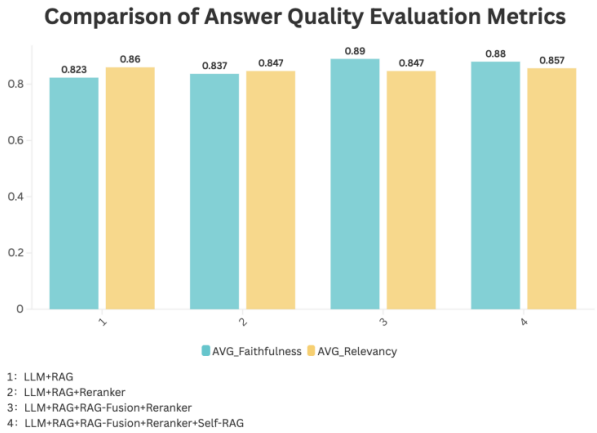

Figure 7. Comparison of Generation Quality

By comparing the evaluation metric data of the four experimental groups, we can draw the following analytical conclusions:

Retrieval Quality Analysis:

In terms of precision, the LLM+RAG+Reranker scheme achieved the best result(0.942), an improvement of 1.3 percentage points over the baseline LLM+RAG scheme. This indicates that the re-ranking module can effectively improve the accuracy of retrieval results.

Recall showed a similar trend, with the LLM+RAG+Reranker scheme reaching 0.938, an improvement of 1.6 percentage points over the baseline scheme. This suggests that re-ranking not only improves accuracy but also expands the coverage of relevant documents.

In terms of MRR, the baseline LLM+RAG scheme performed best(0.576). This interesting phenomenon may be because the complex re-ranking and fusion mechanisms, while improving overall retrieval quality, have a certain impact on the ranking position of the first relevant document.

NDCG remained at a high level(around 0.98) for all schemes, with little difference, indicating that each scheme can effectively rank documents with high relevance in higher positions.

Generated Answer Quality Analysis:

In terms of faithfulness, the two schemes with RAG-Fusion were significantly better than the first two schemes, especially the LLM+RAG+RAG-Fusion+Reranker scheme, which reached 0.89, an improvement of 6.7 percentage points over the baseline scheme. This indicates that the question rewriting and fusion strategies can help the model generate answers that are more faithful to the source documents.

Relevancy showed relatively small differences across the schemes, all remaining between 0.84 and 0.86, indicating that the basic RAG scheme has achieved a good level of answer relevancy.

In the keyword retrieval evaluation, accuracy reached approximately 90%, as assessed by OpenAI's o1-mini model. Minor discrepancies between the reference answers and those generated by the LLM could lead to slight variations in this score. This high accuracy demonstrates the effectiveness of Qwen2.5-Coder in generating SQL statements, thereby ensuring the reliability of the keyword retrieval component.

In summary, the experimental results show that the introduction of each module has its unique contribution to system performance: The combination of BGE-M3 and BGE-Reranker-V2-M3 does improve retrieval quality. The addition of RAG-Fusion, although slightly reducing some retrieval metrics, significantly improves the faithfulness of answers. The introduction of the Self-RAG reflection mechanism achieves an overall balance of various metrics while maintaining high faithfulness. However, it is important to note that these performance improvements come at the cost of increased system response latency. Therefore, in practical applications, a trade-off between performance and efficiency needs to be made based on the specific scenario.

V. Conclusions

This paper introduces an innovative question-answering system for arXiv papers, leveraging the Qwen 2.5 series of open-source LLMs and advanced technologies like LlamaIndex. By demonstrating significant advancements in cross-lingual retrieval and fine-tuned question-answering capabilities, this work provides compelling evidence of the transformative potential of LLMs and RAG in academic applications.

The system's modular design, incorporating intelligent query routing and multilingual re-ranking with the BGE model series, ensures both scalability and adaptability. This enables researchers to pose questions in Chinese and retrieve precise, semantically relevant information from English academic literature, significantly enhancing the accessibility and utility of global scholarly resources. Moreover, the strategic application of prompt engineering throughout the system underscores its critical role in optimizing response quality and format, further demonstrating the practical value of effective LLM-driven design.

Beyond its immediate contributions, this research establishes a foundation for future work in academic question-answering. We show that smaller, fine-tuned LLMs can be a cost-effective solution for institutions with limited resources, enabling them to utilize advanced AI tools. Our future plans include integrating the latest arXiv papers, specializing in areas like RAG, and expanding to vision-language models for visual question-answering. We aim to create a versatile and evolving platform that, through continuous improvement and user feedback, will become a key resource for intelligent academic tools, ultimately driving research efficiency and interdisciplinary innovation.

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